RESEARCH PAPER MANUSCRIPT

Product Optimization and Revenue Contribution Analysis of Afficionado Coffee Roasters Using Interactive Data Analytics

Arul Jothi P.

Data Analytics and Business Intelligence Project
Afficionado Coffee Roasters Case Study

Prepared from the supplied project report and reported dashboard analysis

Abstract

Retail transaction data can help businesses understand product demand, category contribution, and store-level sales patterns. This paper presents a descriptive analytics case study using the Afficionado Coffee Roasters transaction dataset and an interactive dashboard. The supplied project report records 149,116 transaction records, 80 unique products, total revenue of Rs. 698,812.33, and total quantity sold of 214,470 units. Revenue was examined by category, product, and store location, with a Pareto-style cumulative contribution analysis used to explore revenue distribution across products. Coffee and Tea together accounted for approximately 66.74% of reported revenue, while Hell's Kitchen recorded the highest revenue among the three store locations. Python and Pandas were used for data processing, Plotly for visualizations, and Streamlit for dashboard delivery. These results describe the supplied dataset and do not establish profitability or causal explanations because product costs, discounts, and operating expenses were not included in the reported analysis.

Keywords

Keywords: Retail analytics; revenue contribution; product performance; Pareto analysis; Python; Pandas; Plotly; Streamlit; business intelligence.

1. Introduction

Retail businesses generate transaction data that can be transformed into operational information through descriptive analytics. Reviewing revenue by category, product, and store can help decision-makers understand where recorded sales are concentrated and where further investigation may be useful. An interactive dashboard makes these views easier to explore than a static summary alone. This study uses the Afficionado Coffee Roasters transaction dataset as a case study and organizes the reported analysis around overall performance, category contribution, leading products, store comparisons, and cumulative product revenue contribution. The dashboard is an exploratory business intelligence tool, not a forecasting or causal model.

2. Aim and Objectives

The aim is to demonstrate how transaction data can be analyzed and presented in an interactive dashboard to support product and revenue analysis.

Objectives:
• Summarize revenue, sales quantity, transaction count, and product coverage.
• Compare revenue contribution across product categories.
• Identify products with the highest reported revenue.
• Compare revenue across store locations.
• Examine cumulative revenue contribution using a Pareto-style ranking.
• Present the analysis through interactive filters and visualizations.

3. Dataset Description

The project report describes 149,116 transaction records and 11 columns: transaction_id, year, transaction time, transaction_qty, store_id, store_location, product_id, unit_price, product_category, product_type, and product_detail. The report states that zero missing values and zero duplicate rows were reported after checking the dataset.

Transaction revenue was calculated using the formula:

Revenue = Transaction Quantity × Unit Price

The figures in this manuscript are taken from the supplied project report and should be rechecked against the final dataset and deployed dashboard before formal submission or publication.

4. Tools and Method

The workflow described in the project report uses Python and Pandas for data loading, preparation, grouping, and aggregation; Plotly for interactive charts; and Streamlit for dashboard delivery. Revenue contribution percentages were calculated by comparing each group’s revenue with total revenue. Products were ranked by revenue for cumulative contribution analysis. The dashboard includes key performance indicators, category revenue analysis, product comparisons, product quantity-versus-revenue analysis, store comparisons, Pareto analysis, interactive filters, and filtered-data download.

Median-based product segments are descriptive classifications. They do not independently indicate profitability, product quality, or the cause of sales performance.

5. Results

• 1 Overall performance

The project report records total revenue of Rs. 698,812.33, total quantity sold of 214,470 units, 149,116 transaction records, 80 unique products, and average revenue per transaction of approximately Rs. 4.69.

• 2 Category revenue contribution

The reported category figures are summarized in Table 1.

Table 1. Reported revenue by product category

Product category	Revenue (Rs.)	Revenue share
Coffee	269,952.45	38.63%
Tea	196,405.95	28.11%
Bakery	82,315.64	11.78%
Drinking Chocolate	72,416.00	10.36%
Coffee beans	40,085.25	5.74%
Branded	13,607.00	1.95%
Loose Tea	11,213.60	1.60%
Flavours	8,408.80	1.20%
Packaged Chocolate	4,407.64	0.63%

5.3 Top revenue-generating products

The five leading products by revenue in the supplied report were Sustainably Grown Organic Lg (Rs. 21,151.75), Dark chocolate Lg (Rs. 21,006.00), Latte Rg (Rs. 19,112.25), Cappuccino Lg (Rs. 17,641.75), and Morning Sunrise Chai Lg (Rs. 17,384.00). Sustainably Grown Organic Lg was the highest-revenue product in the reported analysis.

5.4 Store-level comparison

The reported store revenues were Hell's Kitchen at Rs. 236,511.17, Astoria at Rs. 232,243.91, and Lower Manhattan at Rs. 230,057.25. Hell's Kitchen recorded the highest revenue among the three locations, although the values were relatively close. Revenue alone cannot explain these differences; product mix, customer traffic, availability, costs, and operating conditions would need separate analysis.

5.5 Pareto-style revenue contribution

The report states that the dataset covers 80 unique products and that approximately 43 products (53.75% of the product count) were needed to reach 80% of cumulative revenue when products were ranked by revenue. In the reported analysis, revenue was distributed across a substantial portion of the product range rather than being limited to a very small number of products.

6. Discussion

Coffee and Tea are the largest revenue-contributing categories in the supplied analysis. Their combined share is approximately 66.74%, so monitoring these categories may help explain changes in overall revenue. The product ranking provides a concise view of high-revenue items, while store comparisons show relatively similar revenue totals across the three locations.

These results are descriptive, not causal. Higher revenue does not necessarily mean higher profit, and a high-volume product with comparatively lower revenue is not necessarily underperforming. Product costs, discounts, margins, inventory availability, seasonality, customer counts, and operating expenses would be needed to support stronger product optimization decisions.

7. Practical Recommendations

• Monitor Coffee and Tea category revenue over time because these categories represent the largest combined share in the report.

• Track availability and sales trends for leading revenue products.

• Investigate high-volume products with lower revenue using product size, price, discount, and product-mix information.

• Compare store performance alongside transaction patterns, customer traffic, product mix, and operating costs.

• Add cost and discount data before drawing conclusions about profitability, pricing, or margin optimization.

• Use consistent date ranges and verified calculations when comparing reporting periods.

8. Limitations

The reported analysis focuses on revenue and quantity rather than net profit. Product cost, operating expenses, and discount information were not included in the project report. Median-based product classification does not establish poor profitability. Pareto analysis describes the distribution in the analyzed dataset and does not guarantee future performance. Metrics and calculations should be independently validated against the final data and deployed dashboard before this manuscript is submitted as formal research.

9. Conclusion

This case study demonstrates how transaction data can be transformed into an interactive business intelligence dashboard using Python, Pandas, Plotly, and Streamlit. The supplied analysis summarizes overall sales, category revenue contribution, leading products, store-level revenue, and cumulative product contribution. The reported results identify Coffee and Tea as the largest combined category contributors and Hell's Kitchen as the highest-revenue store among the three locations.

The dashboard provides a practical interface for exploring these patterns, but results should be interpreted within the limits of descriptive revenue analysis. Adding costs, discounts, customer behavior, and historical trends would enable more complete profitability analysis and stronger product optimization decisions.

References

• Python Software Foundation. Python Documentation. https://docs.python.org/

• pandas development team. pandas documentation. https://pandas.pydata.org/docs/

• Plotly Technologies Inc. Plotly Python Graphing Library. https://plotly.com/python/

• Streamlit, Inc. Streamlit Documentation. https://docs.streamlit.io/

Publication Status Note

This is a research-paper-style manuscript drafted from the supplied project report. It has not been peer reviewed or published by a journal. Before submission, verify all figures against the final dataset, add any required institutional details, and follow the recipient’s requirements for originality, references, and publication status.